NIKHIL RAO

Junior Python Backend Developer

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PROFILE

Computer science graduate with hands-on experience building Flask REST APIs, integrating PostgreSQL, and writing pytest tests. Interested in an entry-level backend role involving readable Python code, debugging, and database-backed services.

TECHNICAL SKILLS

Python: OOP, lists, dictionaries, sets, exception handling, modules, virtual environments and pip. Backend: Flask, REST APIs, HTTP methods, JSON, input validation and SQLAlchemy. Databases: PostgreSQL, SQLite, SQL joins, constraints and transactions. Tools: Git, GitHub, pytest, Postman, Docker, Linux and basic AWS EC2. Other: Pandas, HTML, CSS and basic JavaScript.

INTERNSHIP EXPERIENCE

Python Backend Intern | SampleWorks Labs (fictional) | Jan 2026 - Apr 2026

- Implemented Flask endpoints for an internal task tracker, including creating tasks, listing filtered results and updating task status. Separated routes, service logic and SQLAlchemy database models.
- Validated required fields and due dates; returned JSON errors with appropriate HTTP 400, 404 and 409 responses. Used PostgreSQL foreign keys and transaction rollback to handle invalid updates.
- Wrote pytest tests using fixtures and a separate test database for successful requests, missing fields and duplicate records. Reproduced defects with Postman and reviewed fixes through Git pull requests.

PROJECTS

Course Booking REST API | Python, Flask, PostgreSQL, SQLAlchemy, pytest

- Built endpoints to create courses, list available courses and register learners. Designed Course, Learner and Booking tables with relationships and a unique constraint preventing duplicate bookings.
- Implemented service-layer checks for course capacity and missing records. Used transaction commit/rollback and added pagination for course listings. Tested validation errors, full courses and duplicate bookings.
- Containerized the API with Docker and ran it with PostgreSQL using Docker Compose. Practised deployment on AWS EC2 and connected a simple HTML/JavaScript page to the API using fetch.

CSV Sales Reporting Tool | Python, Pandas, argparse, pytest

- Created a command-line tool to read CSV sales files, validate required columns, remove duplicate rows and summarize monthly revenue by product. Exported results as CSV reports.
- Handled missing files, invalid dates and non-numeric amounts with clear error messages. Added tests using small sample datasets and documented setup and usage commands in the project repository.

EDUCATION

B.E. in Computer Science | Example Institute of Technology (fictional) | 2022 - 2026

Relevant coursework: Data Structures, Object-Oriented Programming, Database Management Systems and Software Engineering.

Fictional candidate profile created solely for capstone application and interview testing.